

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA0301

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO5:** Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Simulation-Based Assessment Question

Duration: 1hr

- 1. Simulate an RL-based smart energy management system for optimizing power distribution in a microgrid.. .**

(5 Marks)

1. Introduction

A microgrid is a small-scale electrical network that can operate independently or together with the main electrical grid. It combines different energy sources such as solar photovoltaic (PV), wind turbines, battery storage, diesel generators, and conventional grid power to supply electricity to consumers. Modern microgrids generate electricity from multiple sources. However, renewable energy sources such as solar and wind are variable. Solar power depends on sunlight, while wind power changes according to weather conditions. Therefore, efficiently distributing power among different sources is an important challenge. Traditional energy management systems generally use predefined rules or optimization methods. These methods may not perform well when the operating conditions change rapidly.

Reinforcement Learning (RL) provides an intelligent solution. An RL agent learns how to make decisions by interacting with the microgrid environment. It observes the current

condition, selects a power-distribution action, receives a reward, and improves its decisions through repeated learning.

Main Objectives

The proposed RL-based system aims to:

- Minimize total operating cost.
- Maximize renewable energy utilization.
- Reduce unnecessary grid dependency.
- Maintain battery charge within safe limits.
- Balance electricity generation and demand.
- Reduce power losses.
- Maintain reliable operation.
- Automatically adapt to changing energy conditions.

Basic Working Principle

Microgrid State



RL Agent



Select Power Distribution



Microgrid Environment



Calculate Reward



Update RL Policy



Repeat

The ultimate objective is to develop an intelligent controller that can determine how much power should be obtained from solar, wind, battery, diesel generator, and the main grid at every time interval.

MICROGRID SYSTEM ARCHITECTURE

2. System Architecture

The proposed microgrid consists of several energy-generation and storage components.

Main Components

1. Solar PV

Solar panels convert sunlight into electrical energy.

Solar generation is generally high during daytime and almost zero at night.

2. Wind Turbine

Wind turbines convert wind energy into electrical power.

Unlike solar energy, wind generation can occur during both day and night, depending on wind speed.

3. Battery Energy Storage System

The battery stores excess electricity and supplies power when generation is insufficient.

Two important battery operations are:

- **Charging:** Excess energy is stored.
- **Discharging:** Stored energy is supplied to the load.

4. Diesel Generator

The diesel generator acts as a backup source when renewable generation and battery power are insufficient.

However, diesel generation has a higher operating cost and produces emissions.

5. Main Grid

The microgrid can purchase electricity from the main grid when required.

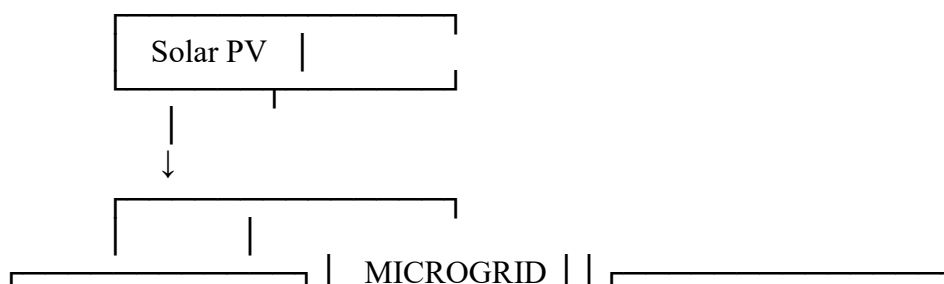
If permitted, excess electricity can also be sold back to the grid.

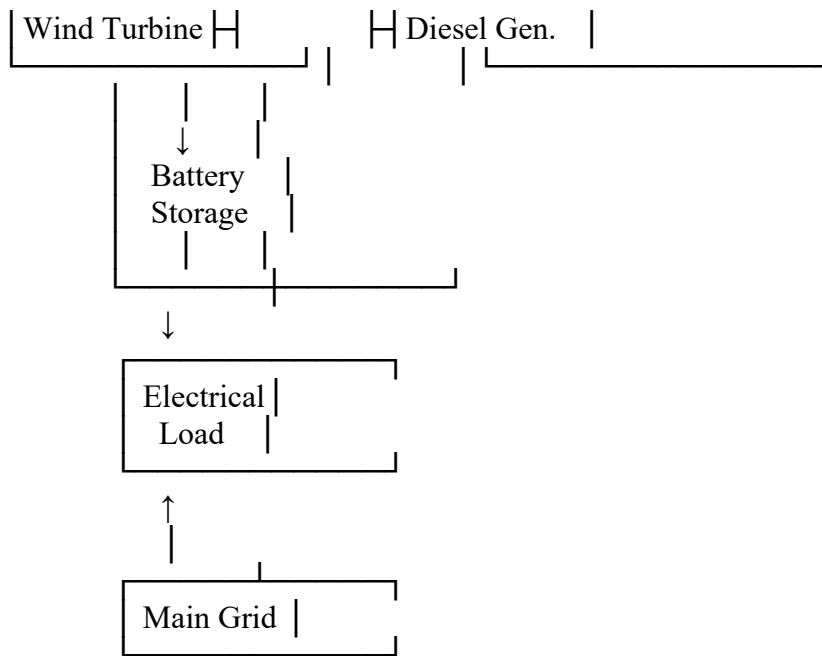
6. Electrical Load

The load represents electricity consumption from:

- Houses
- Buildings
- Industries
- Schools
- Hospitals
- Other electrical equipment

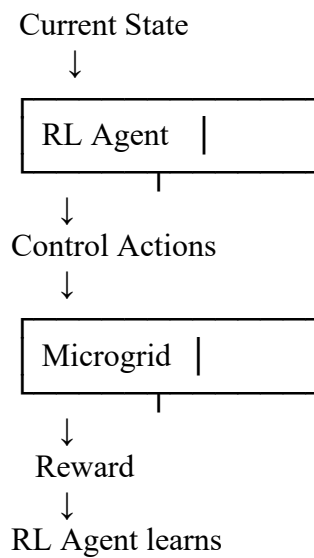
Architecture Diagram





RL Controller

The RL agent sits above the microgrid and determines the power contribution from controllable sources.



STATE SPACE AND ACTION SPACE

3. State Space

The **state** describes the current condition of the microgrid.

A state can be represented as:

$$S_t = [P_{PV}, P_{wind}, P_{load}, S, O, CP_{grid}t]$$

where:

- P_{PV} = available solar power
- P_{wind} = available wind power
- P_{load} = current load demand
- SOC = battery state of charge
- P_{grid} = grid power
- t = time of day

Additional information can also include:

- Electricity price
- Weather conditions
- Battery temperature
- Previous power demand
- Generator status

Example State

Parameter	Example
Solar power	45 kW
Wind power	30 kW
Load	60 kW
Battery SOC	70%
Electricity price	₹0.18/kWh
Time	2 PM

The RL agent uses this information to decide how the available energy should be distributed.

4. Action Space

An **action** represents the decision made by the RL agent.

For example:

$$A_t = [P_{battery}, P_{diesel}, P_{grid}]$$

The agent decides:

- Battery charging/discharging power
- Diesel generator power
- Grid import/export power

Renewable energy is generally used whenever available.

Example

Suppose:

- Solar = 50 kW
- Wind = 20 kW
- Load = 80 kW

The renewable generation is:

$$50 + 20 = 70 \text{ kW}$$

Therefore, the remaining demand is:

$$80 - 70 = 10 \text{ kW}$$

The RL agent may decide to obtain the remaining 10 kW from the battery rather than using the diesel generator.

REWARD FUNCTION AND POWER BALANCE

5. Reward Function

The reward function is one of the most important components of reinforcement learning.

The agent should receive a **high reward for efficient decisions** and a **penalty for undesirable decisions**.

A simplified reward function can be:

$$R_t = -(C_t + \lambda_1 C_{grid} + \lambda_2 C_{battery} + \lambda_3 C_{diesel})$$

where:

- C_t = operating cost
- C_{grid} = grid-related cost
- $C_{battery}$ = battery degradation penalty
- C_{diesel} = diesel operation penalty
- λ = weighting parameters

Desired Behaviour

The reward function should encourage the RL agent to:

- Use renewable energy.
- Reduce grid purchases.
- Avoid unnecessary diesel operation.
- Maintain healthy battery SOC.

- Meet the load demand.
- Minimize overall cost.

6. Power Balance

The microgrid must maintain a balance between generated power and consumed power.

A simplified equation is:

$$P_{PV} + P_{wind} + P_{diesel} + P_{grid} + P_{battery}^{dis} = P_{load} + P_{battery}^{ch} + P_{loss}$$

where:

- P_{PV} = solar generation
- P_{wind} = wind generation
- P_{diesel} = diesel generation
- P_{grid} = grid power
- $P_{battery}^{dis}$ = battery discharge
- P_{load} = electricity demand
- $P_{battery}^{ch}$ = battery charging
- P_{loss} = system losses

The equation ensures that the electricity supplied is sufficient to meet the load.

Example

Suppose:

- Solar = 50 kW
- Wind = 20 kW
- Battery discharge = 10 kW
- Load = 80 kW

Then:

$$50 + 20 + 10 = 80 \text{ kW}$$

Therefore, the load is perfectly satisfied.

REINFORCEMENT LEARNING ALGORITHM

7. RL-Based Energy Management

The proposed system can use **Deep Reinforcement Learning** for handling a continuous energy-control problem.

One suitable algorithm is **Deep Deterministic Policy Gradient (DDPG)**.

DDPG is suitable when the agent must produce continuous control values.

For example:

Battery = -20 kW

Diesel = 10 kW

Grid = 15 kW

Instead of selecting only discrete actions such as ON/OFF, the agent can determine actual power values.

8. DDPG Architecture

DDPG contains two important neural networks:

Actor Network

The actor receives the current state and produces an action.

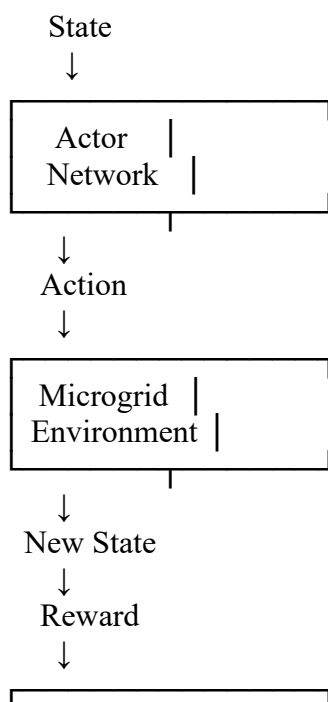
$$A_t = \mu(S_t)$$

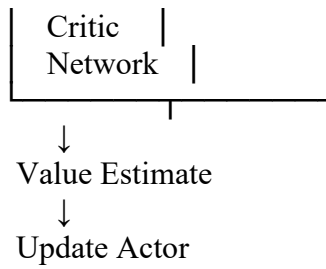
Critic Network

The critic evaluates how good the selected action is.

$$Q(S_t, A_t)$$

Basic Architecture





Learning Process

1. Observe current state.
2. Actor generates an action.
3. Action is applied to the microgrid model.
4. Environment calculates the next state.
5. Reward is calculated.
6. Critic evaluates the action.
7. Actor and critic networks are updated.
8. Process continues for multiple episodes.

SIMULATION SETUP

9. Simulation Environment

A 24-hour microgrid simulation can be created using a time interval such as **15 minutes**.

Therefore:

$$24 \times 4 = 96$$

time steps are used for one day.

Example Simulation Parameters

Parameter	Value
Simulation duration	24 hours
Time interval	15 minutes
Solar capacity	100 kW
Wind capacity	80 kW
Battery capacity	200 kWh
Battery power limit	±60 kW
Diesel capacity	100 kW
Battery efficiency	90%
Discount factor	0.99
Training episodes	500

These values are example simulation parameters and can be changed according to the requirements of the experiment.

10. Assumptions

The simulation can use the following assumptions:

1. Solar, wind, load, and electricity-price profiles are available for the next 24 hours.
2. Solar generation is highest around midday.
3. Solar generation is zero or very low during nighttime.
4. Wind generation changes throughout the day.
5. The battery has minimum and maximum SOC limits.
6. Diesel generation has a higher operating cost.
7. Grid electricity is available whenever necessary.
8. Power demand must always be satisfied.
9. Battery charging and discharging cannot occur beyond its power limit.
10. The RL agent attempts to minimize the total operating cost.

SIMULATION PROCEDURE

11. Simulation Algorithm

Step 1: Initialize Microgrid

Initialize:

- Solar generation
- Wind generation
- Load demand
- Battery SOC
- Electricity prices
- Diesel generator
- Grid connection

Step 2: Initialize RL Agent

Initialize the actor and critic networks.

Step 3: Read Current State

At every time step, the RL agent receives:

$$S_t = [P_{PV}, P_{wind}, P_{load}, S, O, CPrice_t]$$

Step 4: Select Action

The agent produces a power-distribution action.

Step 5: Apply Constraints

The action must satisfy:

- Battery power limits
- Battery SOC limits
- Diesel generator limits
- Grid limits
- Power balance

Step 6: Calculate Power Flow

The system determines how much power comes from:

- Solar
- Wind
- Battery
- Diesel
- Grid

Step 7: Calculate Reward

The system calculates the operating cost and reward.

Step 8: Update RL Agent

The RL agent learns from the state, action, reward, and next state.

Step 9: Continue

The process repeats until the end of the simulation period.

Algorithm

START



Initialize Microgrid



Initialize RL Agent



Read State



Select Action



Apply Action Constraints



Calculate Power Distribution

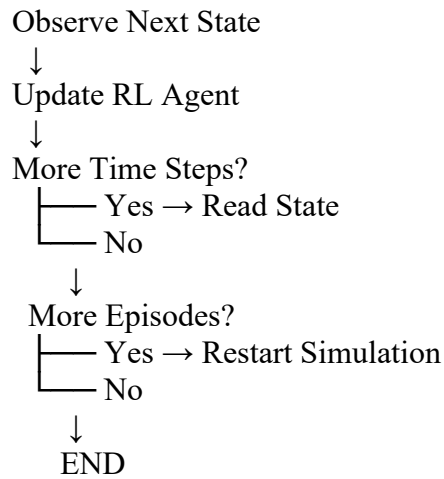


Check Power Balance



Calculate Cost and Reward





SIMULATION RESULTS AND ANALYSIS

12. Expected Simulation Results

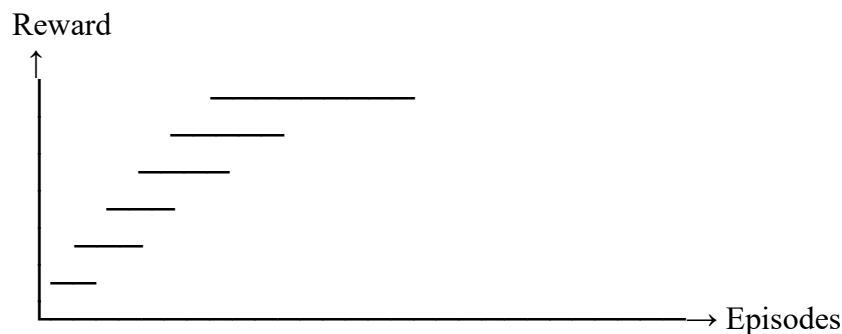
After training, the RL agent should learn to distribute power more efficiently.

The following results can be analyzed:

1. Training Reward

Initially, the RL agent makes poor decisions.

As training progresses, the cumulative reward should generally improve.



This indicates that the agent is learning better power-distribution strategies.

13. Operating Cost

The total operating cost can be compared between:

- Rule-based control
- Conventional optimization
- RL-based control

An effective RL controller should reduce unnecessary expensive energy purchases.

Example:

Method	Daily Cost
Rule-Based Control	₹152.6
Conventional Control	₹119.4
RL-Based EMS	₹87.3

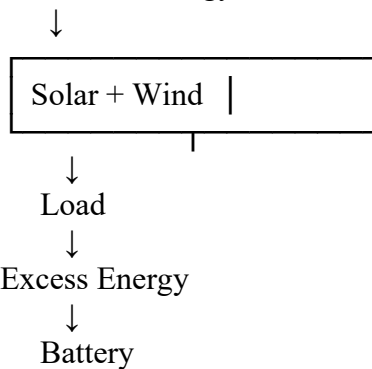
Note: These values are example results for demonstrating how the results section can be presented; actual values should be replaced with the results obtained from your simulation.

4. Renewable Energy Utilization

The RL agent should prioritize renewable sources.

For example:

Renewable Energy

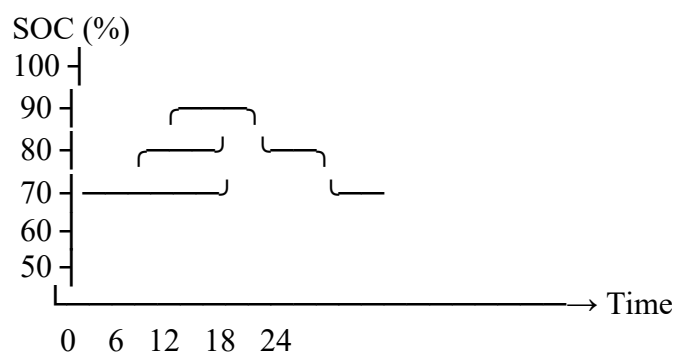


This reduces dependence on diesel generators and grid electricity.

15. Battery SOC Analysis

Battery SOC should remain within safe limits.

Example:



During periods of excess solar or wind generation, the battery charges.

During high-demand periods, the battery discharges.

POWER DISTRIBUTION AND DISCUSSION

16. Power Dispatch

The RL agent should learn a strategy similar to:

Morning

Solar generation begins increasing.

The system uses solar power to supply the load.

Midday

Solar generation is high.

The load is supplied primarily using renewable energy.

Excess solar power can charge the battery.

Evening

Solar generation decreases while demand may increase.

The battery can discharge to reduce grid dependency.

Night

Solar generation is unavailable.

The system may use:

- Battery
- Wind
- Grid
- Diesel backup

depending on availability and cost.

17. Example Power Dispatch

Time	Main Energy Source

00–06	Wind + Grid
06–09	Solar + Wind
09–16	Solar + Wind + Battery Charging
16–19	Battery + Wind
19–22	Battery + Grid

This demonstrates how the RL agent can dynamically change the power distribution according to energy availability and demand.

18. Discussion

The simulation demonstrates several important benefits.

Reduced Operating Cost

The RL agent learns to select lower-cost energy sources whenever possible.

Better Renewable Integration

Solar and wind power are prioritized when available.

Improved Battery Utilization

The battery stores excess renewable energy and supplies power during high-demand periods.

Reduced Grid Dependency

The system can reduce unnecessary electricity purchases from the main grid.

Reduced Diesel Usage

Diesel generation is used mainly as a backup source.

Adaptive Decision-Making

Unlike fixed rule-based systems, the RL agent can learn from changing operating conditions.

CONCLUSION AND FUTURE SCOPE

19. Conclusion

The proposed **RL-based Smart Energy Management System** demonstrates how reinforcement learning can be applied to optimize power distribution in a microgrid.

The RL agent continuously observes the microgrid state, including renewable generation, load demand, battery SOC, electricity price, and time.

Based on this information, it determines the appropriate power-distribution action.

The main objectives are to:

- Minimize operating cost.
- Maximize renewable energy utilization.
- Reduce grid dependency.

- Reduce diesel generator usage.
- Maintain battery health.
- Satisfy electricity demand.
- Improve overall microgrid efficiency.

The use of **Deep Reinforcement Learning**, such as DDPG, is particularly useful when power-control actions are continuous.

The complete system can be summarized as:

<i>State</i> → <i>RL Agent</i> → <i>Action</i> → <i>Microgrid</i> → <i>Reward</i> → <i>Learning</i>

Therefore, reinforcement learning provides an intelligent and adaptive approach for future smart-grid and microgrid energy management systems.

20. Future Scope

The system can be improved in several ways.

1. Real-Time Data

Real-time solar, wind, load, and electricity-price data can be integrated.

2. Weather Prediction

Weather forecasting can improve solar and wind power prediction.

3. Multiple Microgrids

The system can be extended to coordinate multiple interconnected microgrids.

4. Electric Vehicles

Electric vehicles can be included as additional energy-storage resources.

5. Advanced RL Algorithms

Algorithms such as:

- PPO
- SAC
- TD3
- Multi-Agent RL

can be compared with DDPG.

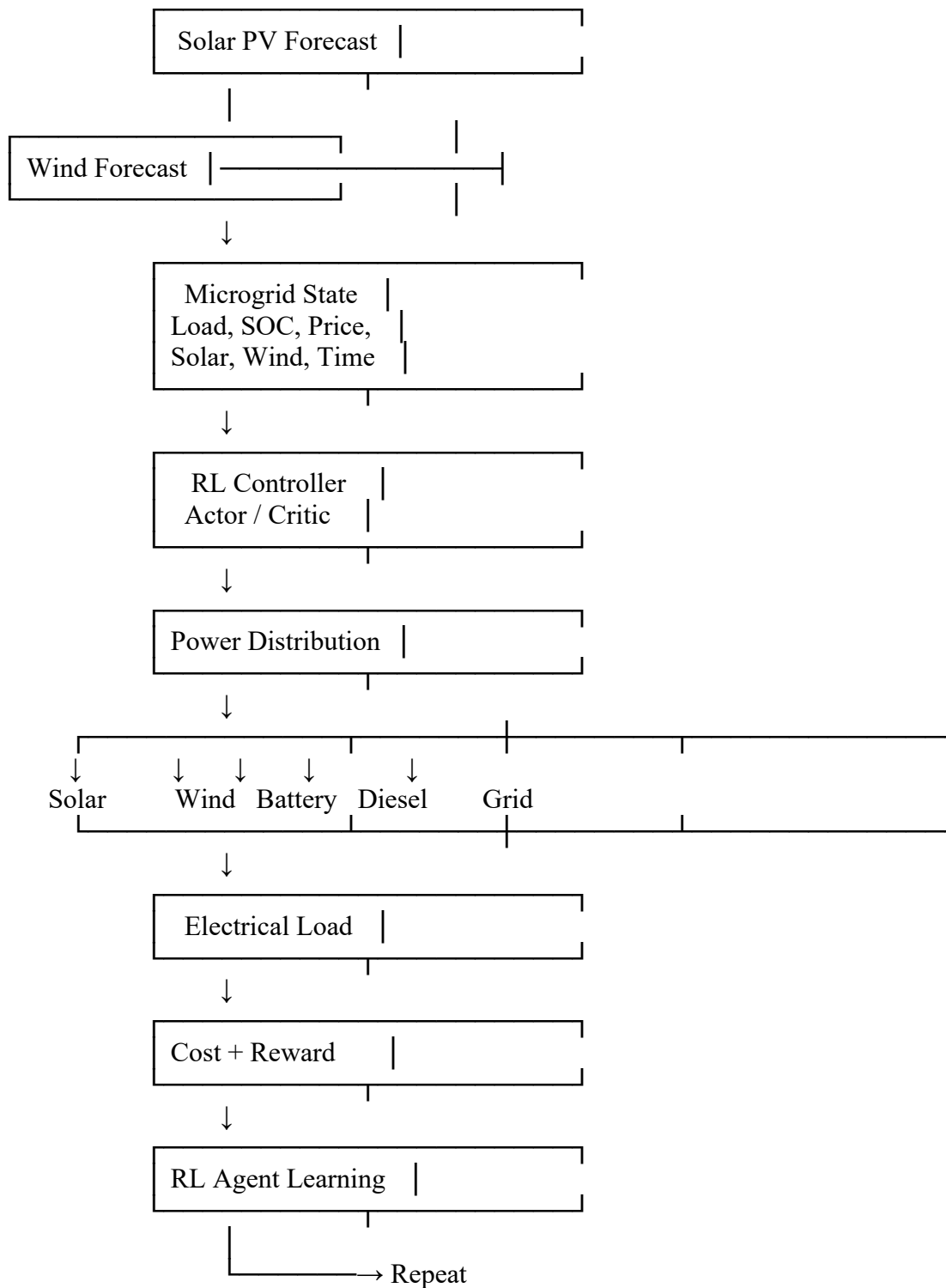
6. Hardware Implementation

The trained controller can eventually be connected to real microgrid hardware or a hardware-in-the-loop simulator.

7. Carbon Emission Optimization

The reward function can include carbon emissions along with operating cost.

21. Final Overall Architecture



Key Terms for Viva

Term	Meaning
Microgrid	Small-scale power network containing multiple energy sources
RL	Learning through interaction and rewards
Agent	Decision-making component
Environment	The microgrid being controlled
State	Current condition of the microgrid
Action	Power-distribution decision
Reward	Feedback showing how good the decision was
Policy	Strategy used to select actions
SOC	Battery State of Charge
DDPG	Deep RL algorithm for continuous actions
Power Balance	Matching generation with demand
EMS	Energy Management System
Renewable Energy	Solar and wind energy
Grid	External electricity network

Code of Conduct:

*I Malli Chandana(192472250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Malli Chandana

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively